

1	The table below shows the estimates of some physical quantities. Which row is not a reasonable estimate?		
		quantity	value
	A	current used in an electric kettle	8 A
	B	mass of an adult male	70 kg
	C	speed of an Olympic sprint runner	10 m s^{-1}
	D	water pressure at the bottom of a public swimming pool	10^4 Pa

Answer: D

Option A: With $I = 8 \text{ A}$, and $V = 240 \text{ V}$, power rating of kettle = $8 \times 240 = 1920 \text{ W}$ which is a reasonable estimate.

Option B: Typically Singaporean adult weighs between 50 kg and 100 kg. Therefore 70 kg is a reasonable estimate.

Option C: The timing for 100 m race in the Olympic is about 10 s; therefore the average speed of 10 m s^{-1} is a reasonable estimate.

Option D: Atmospheric pressure = 10^5 Pa .

Assume that the swimming pool has depth of 2 m,

$$\begin{aligned}
 \text{Pressure at the bottom pool} &= \text{Atmospheric pressure} + h\rho g \\
 &= 10^5 + (2 \times 1000 \times 9.81) \\
 &= 10^5 + 1.96 \times 10^4 = 1.196 \times 10^5 \text{ Pa}
 \end{aligned}$$

10^4 Pa is less than $1.196 \times 10^5 \text{ Pa}$. Therefore 10^4 Pa is not a reasonable estimate. Answer is D.